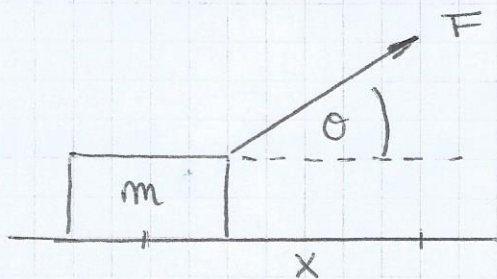
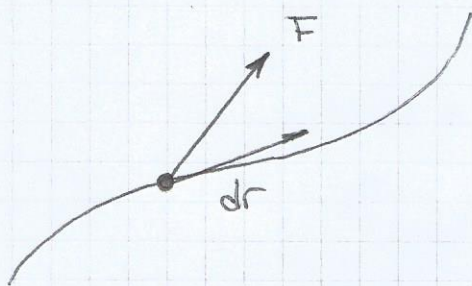


Unidad VII: Trabajo y energía.Trabajo mecánico (esfuerzo):

- F
- Δx
- θ



$$dW = \mathbf{F} \cdot d\mathbf{r} = F \cdot dr \cdot \cos \theta$$

- escalar
- el signo depende del ángulo

$\theta < \pi/2 \Rightarrow dW > 0$ se entrega trabajo al sistema

$\theta = \pi/2 \Rightarrow dW = 0$ no realiza trabajo.

$\pi > \theta > \pi/2 \Rightarrow dW < 0$ se extrae trabajo al sistema.

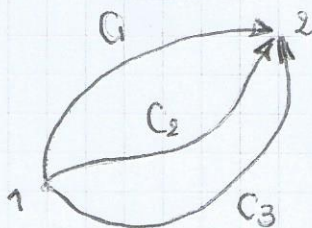
$$W_{1,2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r} \quad \mathbf{F} = F(x, y, z, t) = F_x \hat{x} + F_y \hat{y} + F_z \hat{z}$$

$$d\mathbf{r} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

$$\mathbf{F} \cdot d\mathbf{r} = (F_x \hat{x} + F_y \hat{y} + F_z \hat{z})(dx \hat{x} + dy \hat{y} + dz \hat{z})$$

$$\mathbf{F} \cdot d\mathbf{r} = F_x dx + F_y dy + F_z dz$$

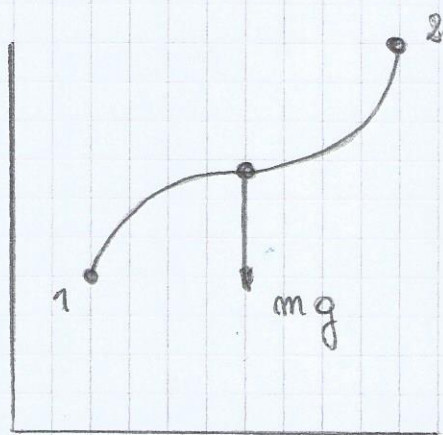
$$\int_{x_1}^{x_2} F_x dx + \int_{y_1}^{y_2} F_y dy + \int_{z_1}^{z_2} F_z dz$$



$$W_{12}^{C1} \neq W_{12}^{C2} \neq W_{12}^{C3}$$

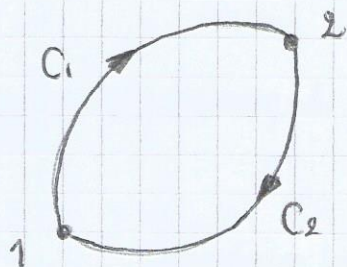
- Las fuerzas conservativas (dependen de la posición), son aquellas en las cuales $W_{12}^{C_1} = W_{12}^{C_2} = W_{12}^{C_3}$

Fuerza Peso:



$$W_{12} = \int_1^2 -mg \hat{z} (dx \hat{x} + dy \hat{y} + dz \hat{z})$$

$$W_{12} = \int_{z_1}^{z_2} -mg dz = -mg(z_2 - z_1)$$



$$C = C_1 + C_2$$

$$W = W_{12}^{C_1} + W_{12}^{C_2}$$

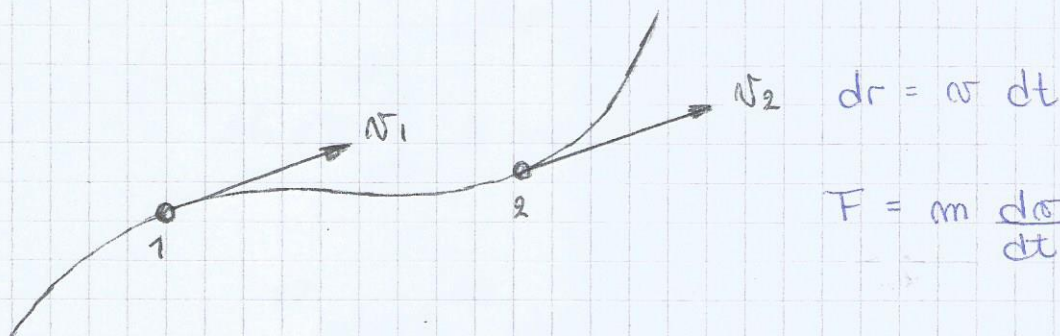
$$W = \int_1^2 dw^{C_1} + \int_2^1 dw^{C_2}$$

$$W = W_{12} - W_{12} = 0 \text{ (F es conservativa)}$$

F conservativa $\oint dw = 0$

Energía: (Capacidad de un sistema de realizar trabajo)

Energía mecánica: $\left\{ \begin{array}{l} \text{movimiento} \Rightarrow \text{Energía cinética} \\ \text{posición} \Rightarrow \text{Energía potencial} \end{array} \right.$



$$dr = v dt$$

$$F = m \frac{dv}{dt}$$

$$W_{12} = \int_1^2 \vec{F} d\vec{r} = \int_1^2 m \frac{d\vec{v}}{dt} \cdot d\vec{r}$$

$$= \int_1^2 m \vec{v} d\vec{v} = \frac{1}{2} \int_1^2 m d(v^2)$$

$$W_{12} = \frac{1}{2} m (v_2^2 - v_1^2)$$

$$\text{Si } v_2 = 0 \Rightarrow W_{12} < 0$$

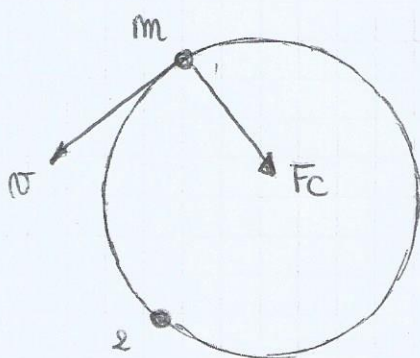
$$W_{12} = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$$

se entrega trabajo

$$E_c = \frac{1}{2} m v^2 \text{ (Energía cinética)}$$

$$W_{12} = E_{c2} - E_{c1}$$

"Teorema de las fuerzas vivas"

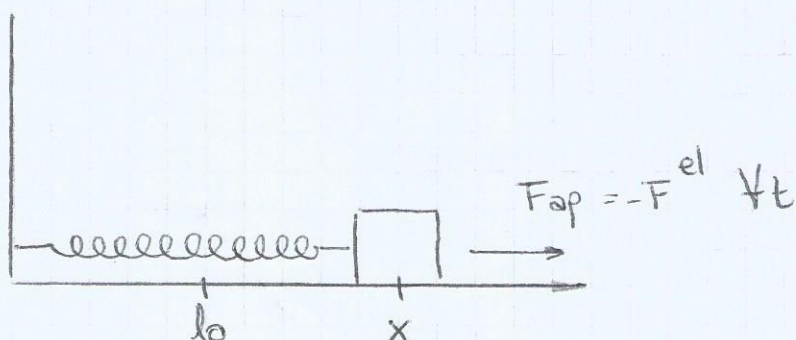


$$\vec{F}_c \perp d\vec{r} \quad W_{12} = \int_1^2 \vec{F}_c d\vec{r} = 0$$

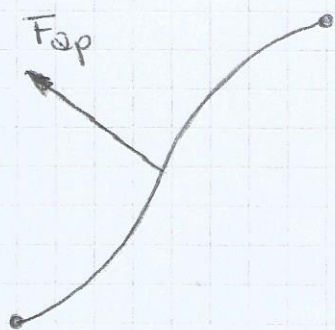
$$W_{12} = 0 \Rightarrow E_{c1} = E_{c2}$$

$$v_1 = v_2$$

Energía potencial (se define para fuerzas conservativas)



$$W = \int_{x_0}^x F_{sp} dx = - \int_{x_0}^x F^{el} dx$$



$$W_{\text{op}} = \int_1^2 F_{\text{op}} dr = - \int_1^2 F_c dr$$

$$W_{\text{op}} = E_p(2) - E_p(1)$$

$$W = \int_{r_0}^r F^{\text{CONS}} dr = - [E_p(r) - E_p(r_0)]$$

$$E_p(r) = - \int_{r_0}^r F^{\text{CONS}} dr + E_p(r_0)$$

$$E_p(r_2) - E_p(r_1) = - \int_{r_0}^{r_2} F^{\text{CONS}} dr + E_p(r_0) - \left[- \int_{r_0}^{r_1} F^{\text{CONS}} dr + E_p(r_0) \right]$$

$$= - \int_{r_0}^{r_2} F^{\text{CONS}} dr + \int_{r_0}^{r_1} F^{\text{CONS}} dr$$

$$= - \int_{r_0}^{r_2} F^{\text{CONS}} dr - \int_{r_1}^{r_0} F^{\text{CONS}} dr$$

$$= - \int_{r_1}^{r_2} F^{\text{CONS}} dr$$

E_p gravitatoria cerca de la superficie de la Tierra

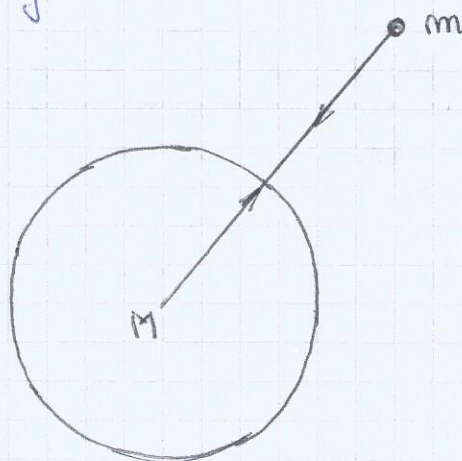
$$W(\text{Peso}) = -m g (z_2 - z_1) \quad \text{Si } z_2 = z$$

$$z_1 = z_0$$

$$E_p(z) = m g (z - z_0) + E_p(z_0)$$

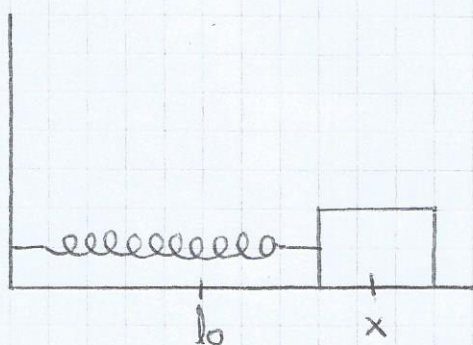
$$\text{Si } E_p(z_0) = 0 \Rightarrow E_p(z) = m g (z - z_0)$$

$$F_g = - \frac{G M m}{r^2} \hat{r}$$



$$\begin{aligned}
 E_p(r) &= - \int_{r_0}^r -\frac{G M m}{r^2} \hat{r} \, dr \hat{r} + E_p(r_0) \\
 &= \int_{r_0}^r \frac{G M m}{r^2} \, dr + E_p(r_0) \\
 &= -\frac{G M m}{r} \Big|_{r_0}^r + E_p(r_0) \\
 &= -\frac{G M m}{r} + \frac{G M m}{r_0} + E_p(r_0)
 \end{aligned}$$

Si $E_p(r_0) = -\frac{G M m}{r_0} \Rightarrow E_p(r) = -\frac{G M m}{r}$



$$F^{el} = -k(x - l_0) \hat{x}$$

$$\begin{aligned}
 E_p(x) &= - \int_{x_0}^x -k(x - l_0) \hat{x} \, dx \hat{x} + E_p(x_0) \\
 &= \int_{x_0}^x k(x - l_0) \, dx + E_p(x_0) \\
 &= \frac{1}{2} k x^2 \Big|_{x_0}^x - k l_0 (x - x_0) + E_p(x_0)
 \end{aligned}$$

Si $x_0 = l_0$ y $E_p(l_0) = 0$

$$E_p(x) = \frac{1}{2} k x^2 - \frac{1}{2} k l_0^2 - k l_0 x + k l_0^2 + 0$$

$$E_p(x) = \frac{1}{2} k (x - l_0)^2$$

$$\text{Si } E_p(r_0) = -\frac{G M m}{r_0}$$

$$E_p(r) = - \int_{r_0}^r F^{\text{cons}} dr + E_p(r_0)$$

$$E_p(r) - E_p(r_0) = \int_{r_0}^r dE_p$$

$$- \int_{r_0}^r F dr = - \int_{x_0}^x F_x dx - \int_{y_0}^y F_y dy - \int_{z_0}^z F_z dz$$

$$\int_{r_0}^r dE_p = \int_{x_0}^x \frac{dE_p}{dx} dx + \int_{y_0}^y \frac{dE_p}{dy} dy + \int_{z_0}^z \frac{dE_p}{dz} dz$$

$$F_x \hat{x} = - \frac{dE_p}{dx} \hat{x}$$

$$F_y \hat{y} = - \frac{dE_p}{dy} \hat{y}$$

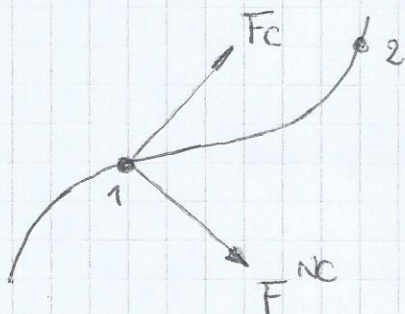
$$F_z \hat{z} = - \frac{dE_p}{dz} \hat{z}$$

$$\text{Si } E_p(z) = m g (z - z_0)$$

$$\text{Si } z=0 \Rightarrow E_p=0$$

$$F_x = - \frac{dE_p}{dz} = -mg$$

$$F_z \hat{z} = -mg \hat{z}$$



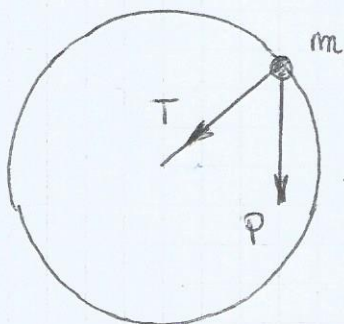
$$W_{12} = W_{12}^c + W_{12}^{nc}$$

$$W_{12} = - [E_p(2) - E_p(1)]$$

$$W_{12} = E_c(2) - E_c(1) \quad (\text{por TFV})$$

Energía mecánica: $E_m = E_c + E_p$

$$E_m(2) - E_m(1) = W_{12}^{NC} \quad \text{Si } W_{12}^{NC} = 0 \Leftrightarrow E_m = \text{cte}$$



$$dW^T = T dr$$

$$dW^T = T r d\theta$$

$$r \perp T \Rightarrow dW^T = 0 \Rightarrow E_m = \text{cte}$$

(La tensión es no conservativa)

Unidades:

C.g.s

M.K.S

ergio

joule

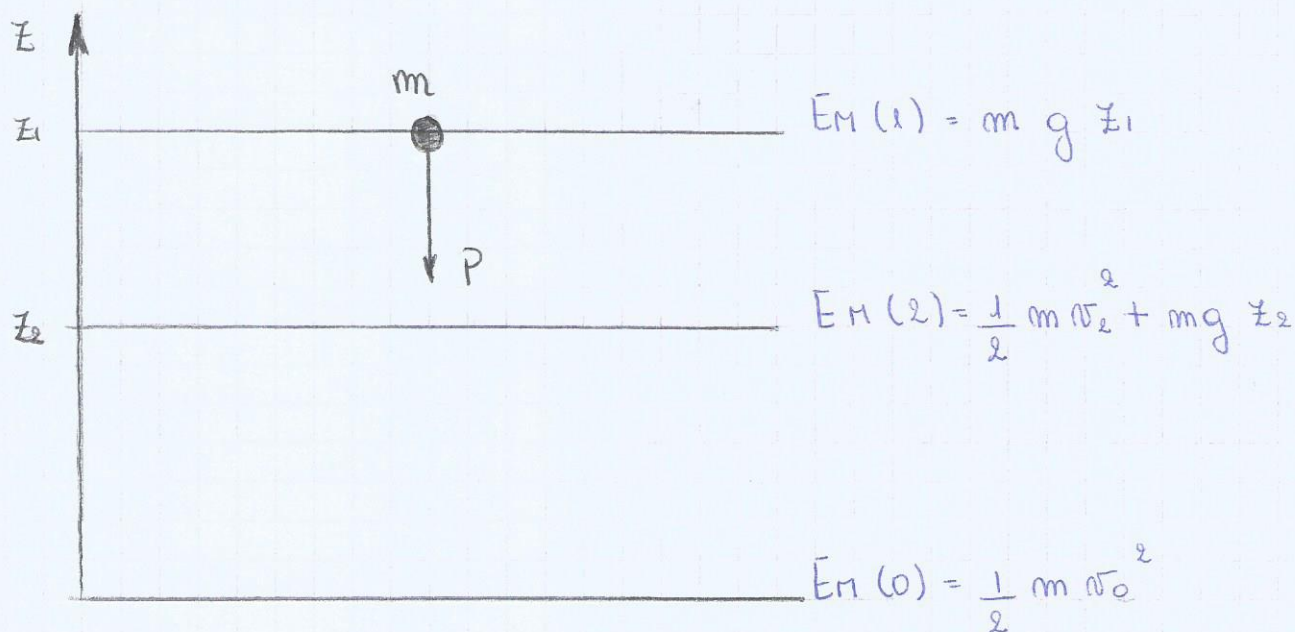
(erg)

J

(dyn. cm)

(N. m)

$$1 \text{ J} = 10^7 \text{ erg.}$$



$$E_m(1) = E_m(2)$$

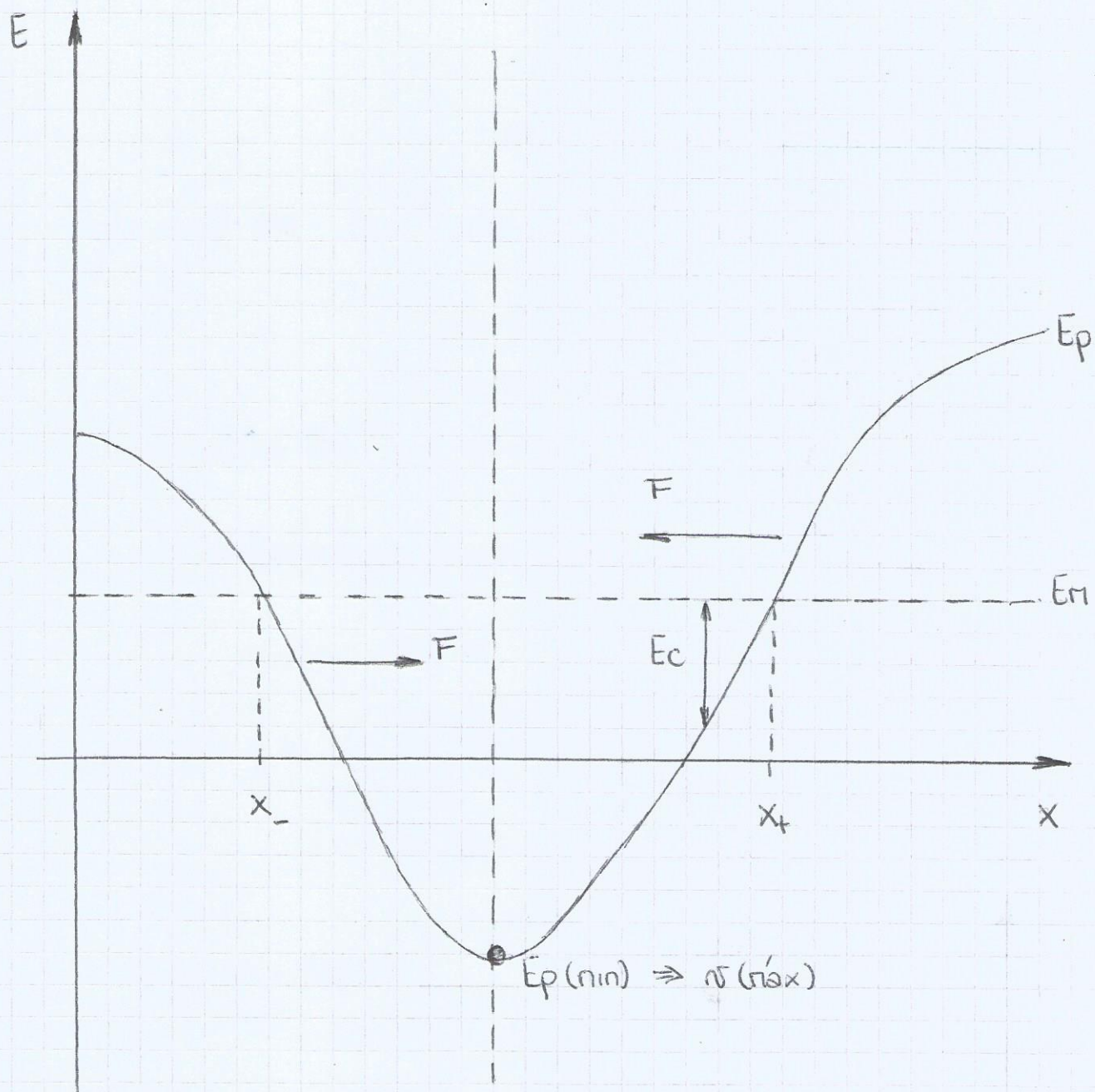
$$m g z_1 = \frac{1}{2} m v_2^2 + m g z_2$$

$$v_2 = \sqrt{2 g (z_1 - z_2)} \quad \hat{z}$$

$$E_m(0) = E_m(1)$$

$$\frac{1}{2} m v_0^2 = m g z_1$$

$$v_0 = \sqrt{2 g z_1} \quad \hat{z}$$



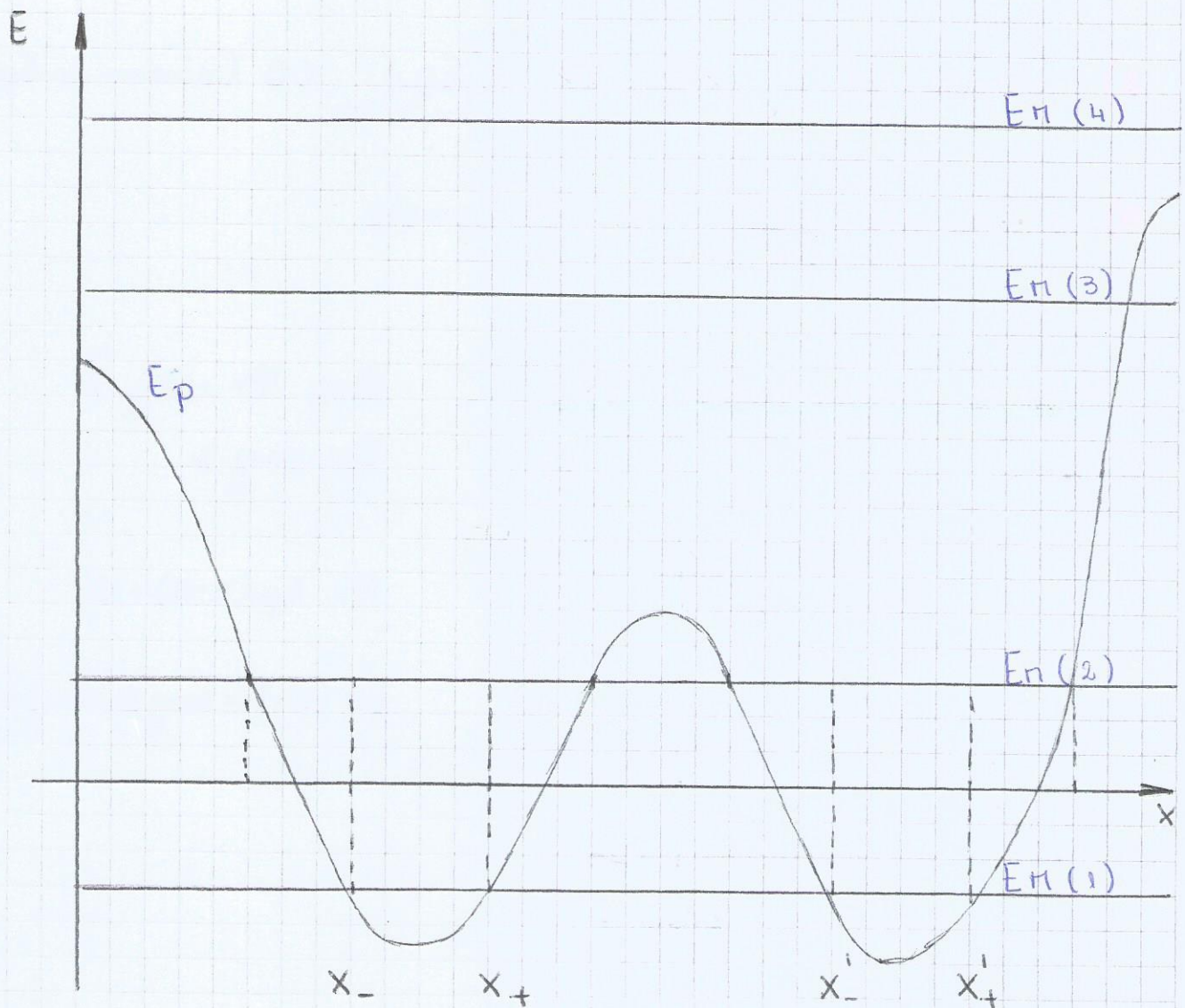
Movimiento ligado entre los puntos de retorno $x^\pm$

Solo hay movimiento si $E_m \geq E_p$

$$E_m = E_c + E_p$$

$$E_c = E_m - E_p \geq 0$$

Pozo de potencial



$$E_{n1} \Rightarrow x_- \leq x \leq x_+ \quad \text{o} \quad x'_- \leq x \leq x'_+$$

$$E_{n2} \Rightarrow x''_- \leq x \leq x''_+$$

$$E_{n3} \Rightarrow -\infty \leq x \leq x'''_+$$

$$E_{n4} \Rightarrow -\infty \leq x \leq \infty \quad (\text{movimiento no ligado})$$

Por definición $F(x_e) = 0$ y $F = -\frac{dE_p}{dx} \hat{x}$

$$E_p(x \approx x_e) = E_p(x_e) + \left. \frac{dE_p}{dx} \right|_{x_e} (x - x_e) - \frac{1}{2} \left. \frac{d^2 E_p}{dx^2} \right|_{x_e} (x - x_e)^2$$

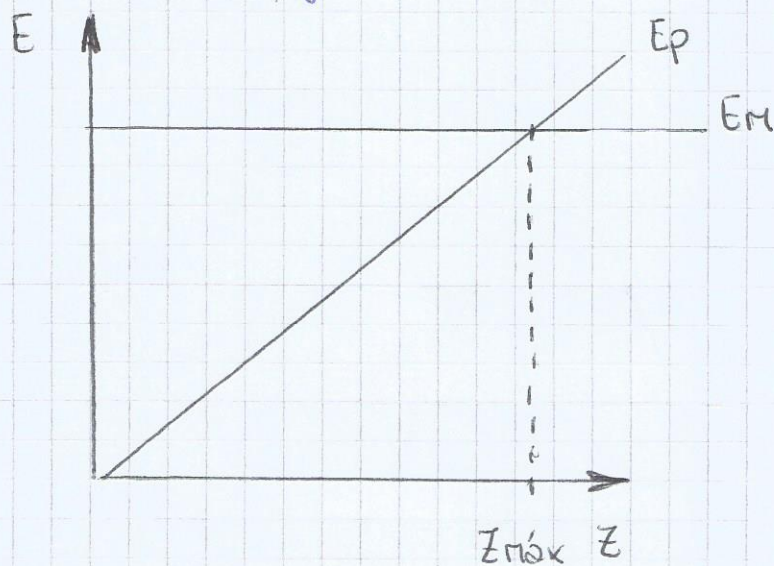
Si $\left. \frac{dE_p}{dx} \right|_{x_e} = 0 \Rightarrow x_e$ es un extremo de E_p

1) Estable $\left. \frac{dF}{dx} \right|_{x_e} < 0 \Leftrightarrow \left. \frac{d^2 E_p}{dx^2} \right|_{x_e} > 0$ (mínimo de E_p)

2) Inestable $\left. \frac{dF}{dx} \right|_{x_e} > 0 \Leftrightarrow \left. \frac{d^2 E_p}{dx^2} \right|_{x_e} < 0$ (máximo de E_p)

3) Indiferente $\left. \frac{dF}{dx} \right|_{x_e} = 0 \Leftrightarrow E_p = \text{cte}$

Potencial gravitatorio (P)



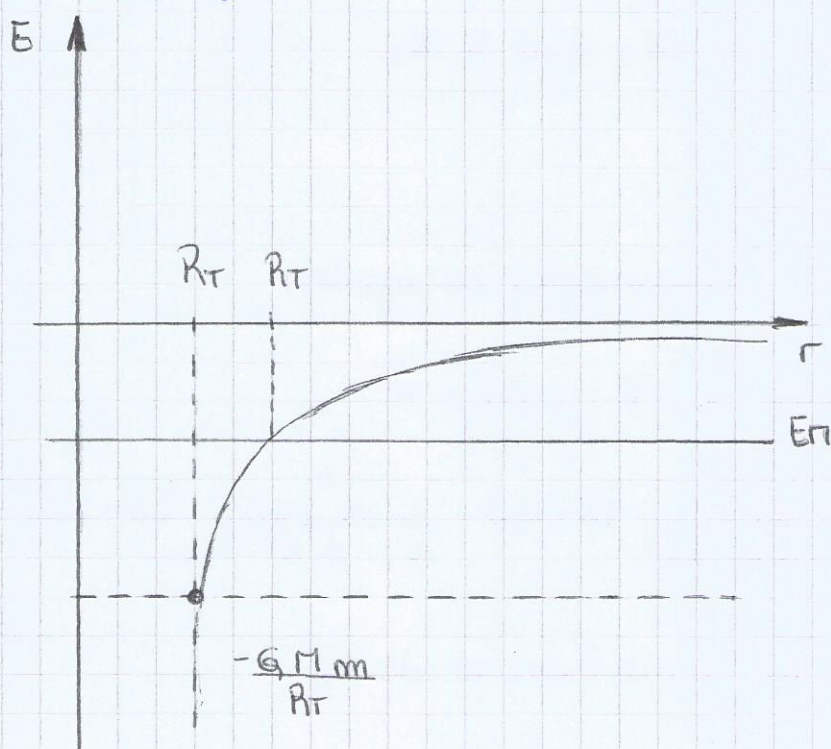
Sea $P = -mg \hat{z}$

$E_p = mgz$

Si $E_p(z=0) = 0$

$E_m(z_m) = mgz_{\text{máx}} = \frac{1}{2} m v_0^2$

Fuerza gravitatoria (Fg):



Si $E_m = 0$, se va al infinito con $v=0$
(velocidad de escape)

Si $E_m > 0$ se va al infinito con v

$$\text{Sea } F_g = -\frac{G M m}{r^2} \hat{r}$$

$$E_p = -\frac{G M m}{r}$$

$$\text{Si } E_p(r_0) = -\frac{G M m}{r}$$

$$E_m = \frac{1}{2} m v^2 - \frac{G m M}{r}$$

$$0 = \frac{1}{2} m v_e^2 - \frac{G m M}{r}$$

$$v_e = \pm \sqrt{\frac{2 G M}{R_T}}$$